

Major Revision: Pairing Proofs and Polynomial Ring Toolkit

Submission: TCHES 2027, Issue 1, Paper #185

Original submission: TCHES 2026, Issue 4, Paper #104

All modified or added parts are highlighted in yellow in the revised paper.

Summary of changes

Location	Change	Criteria
Abstract, §1.2, §2.1	Updated headline numbers: uniform 3-pairing benchmark, BLS12-381 added	A.2, A.3
§3	Concrete Fel23 comparison; new Garaga paragraph	A.1
§4.1, §4.2	Sparseness statement; fixed benchmark cross-reference sentence	A.3, A.6
Algorithms 3, 4	Inlined PolyMulDiv/EvalPoly signatures (two floats removed)	A.6
§5.5.1	Improved Fiat-Shamir section removed redundant Non Interactive figure	A.4
§5.5.2 (new)	Random-oracle instantiations with per-scheme commitment costs across CairoVM, gnark Groth16 and gnark PLONK	A.5
§5.6	Standard indeterminate notation applied	A.6
§6.3	Removed excessive API documentation, added description of the logic flow with algorithm to implementation mappings	A.4
§6.5	Dynamic overflow-tracking note	—
§7.2	Before/After & baseline definitions, metrics paragraph, rewritten benchmark circuit; Table 7 regenerated (BLS12-381, uniform circuit)	A.2, A.3, A.5

Revision criteria

Updates for post-rebuttal discussion requirements,

A.1 Clarify the main contribution relative to the most closely related prior works (such as Fel23 and Garaga)

- Section 3 now quantifies the delta over [Fel23] concretely and links to Sections 4.1 for more details.
- New closing paragraph of Section 3 clarifies Garaga relationship.

A.2 Add additional benchmarks for BLS12-381, as per the rebuttal

- The toolkit is now implemented for BLS12-381 and Table 7 includes BLS12-381 results for both SCS and R1CS.

A.3 Clarify the benchmarking setup, in particular the 3-pairing setting

- We replaced the original composite Groth16 simulation circuit with a uniform 3-pairing product check, $\prod_{i=1}^3 e(P_i, Q_i) = 1$ via **PairingCheck**. The simplified circuit now directly corresponds to the polynomials specified in section 4. Baseline and our work fork are both benchmarked on the same circuit. All numbers regenerated (abstract, Section 1.2, Table 7).

- New text in Section 7.2 states the Miller line sparseness; Section 4.1 now clarifies the sparseness savings are included all compared schemes.

A.4 Improve the mapping between algorithms and implementation, in particular the application of Fiat-Shamir

- Section 5.5 (Fiat-Shamir Transformation) now more detailed, The redundant non-interactive protocol figure was removed.
- Section 6.3 (Algorithms and Implementation) now traces Algorithms 3/4 (and steps in the proof) to the code.

A.5 Qualify the commitment-based performance metric: schemes, overhead, impact

- New Section 5.5.2 (Instantiating the Random Oracle): Details the commitments machinery for CairoVM and gnark(both Groth16 and PLONK).
- Commitment definitions tightened, clear distinction between commitment count (2 commitments) and number of elements committed.

A.6 Minor presentation issues pointed out by the reviewers

- Percentages only included in text in the abstract and the introduction.
- Broken sentences/spelling mistakes fixed
- Formal indeterminate is now capital X throughout the paper.